

Module - 3

Techniques of Counting

- *Basic Counting Principles
 - *Permutations
 - *Combinations
 - *Pigeonhole Principle
- *Inclusion
 - *Exclusion Principle.

Basic Counting Principle:

THE PRODUCT RULE Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and for each of these ways of doing the first task, there are n_2 ways to do the second task, then there are $n_1 n_2$ ways to do the procedure.

THE SUM RULE If a task can be done either in one of n_1 ways or in one of n_2 ways, where none of the set of n_1 ways is the same as any of the set of n_2 ways, then there are $n_1 + n_2$ ways to do the task.

Problems:

A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from?

Solution: The student can choose a project by selecting a project from the first list, the second list, or the third list. Because no project is on more than one list, by the sum rule there are $23 + 15 + 19 = 57$ ways to choose a project. 

① How many different 8 bit strings are there

i) begin ~~&~~ end with one ii) end with 011

iii) begin with 01 & end with 01

Sol

$$\frac{01}{2^1} \frac{01}{2^1} \frac{01}{2^1} \frac{01}{2^1} \frac{01}{2^1} \frac{01}{2^1} \frac{01}{2^1} \frac{01}{2^1}$$

i) Begin with one & end with one

$$\frac{1}{2^0} + \frac{001}{2} + \frac{001}{2} + \frac{01}{2} + \frac{01}{2} + \frac{01}{2} + \frac{01}{2} + \frac{1}{2^0}$$

$$= 2^6$$

either 0
(or) 1

= 1 m/-
of way

$$= 2^0$$

ii) end with 0111

$$\underbrace{0(0r)1}_{2^1} \times \underbrace{0(0r)1}_{2^1} \times \underbrace{0(0r)1}_{2^1} \times \underbrace{0(0r)1}_{2^1} * \underbrace{0}_{2^0} + \underbrace{1}_{2^0} + \underbrace{1}_{2^0} + \underbrace{1}_{2^0}$$

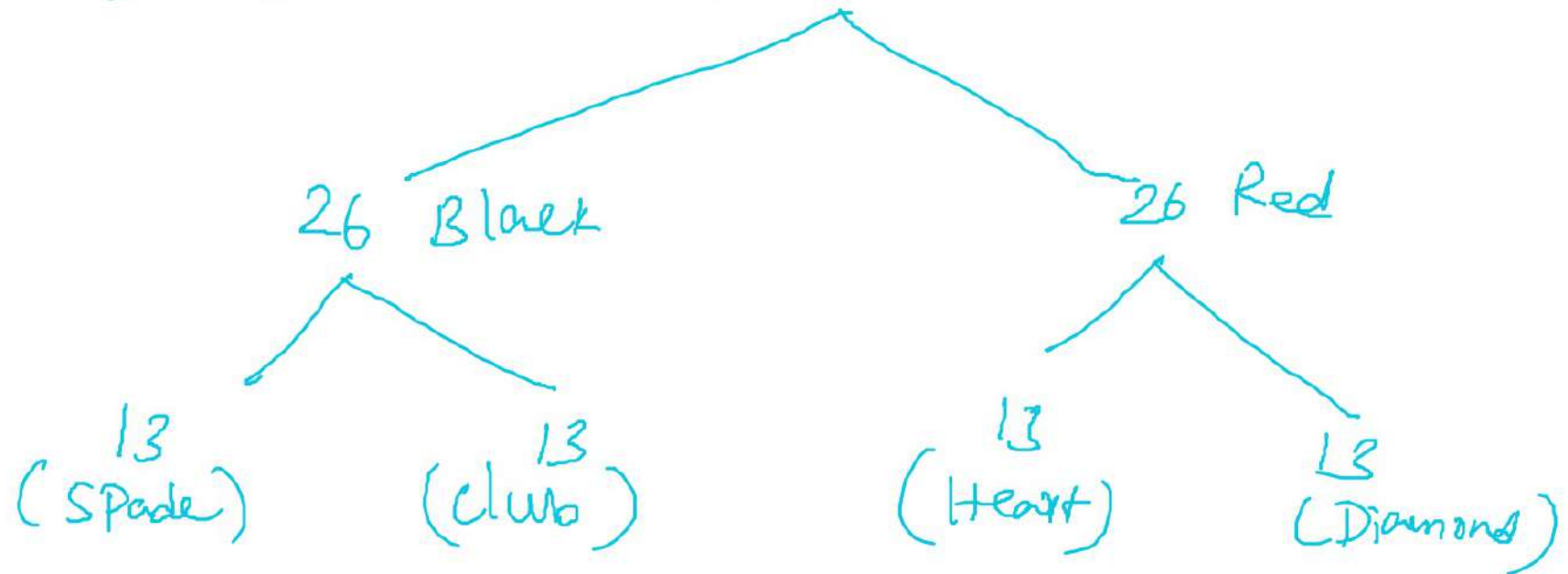
$$= 2^4 = 16$$

iv) begin with 01 & end with 01

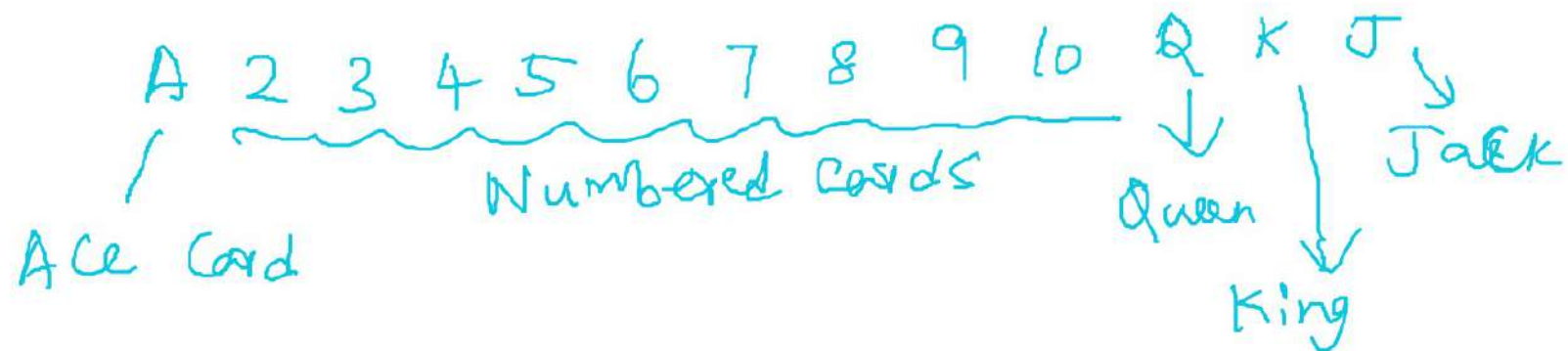
$$\underbrace{0}_{2^0} \times \underbrace{1}_{2^0} \times \underbrace{0(0r)1}_{2^1} \times \underbrace{0(0r)1}_{2^1} \times \underbrace{0(0r)1}_{2^1} \times \underbrace{0(0r)1}_{2^1} \times \underbrace{0}_{2^0} \times \underbrace{1}_{2^0}$$

$$= 2^4 = 16$$

In Card Game : 52 Card



In each - 13



- ② In how many ways can we draw
- i) a heart (or) a Spade from an ordinary deck of playing card?
 - ii) A heart (or) an ace?
 - iii) An ace (or) a king?
 - iv) A card numbered 2 through 10?

Sol

i) $13 + 13 = 26$ ways

ii) $13 + 3 = 16$ ways

iii) $4 + 4 = 8$ ways

iv) $9 + 9 + 9 + 9 = 36$ ways

$$\begin{array}{cccccccc} 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline & & & & & & & 10 \\ & & & & & & 9 & \end{array}$$

③ How many different license plates are available? Each plate contains a sequence of 3 letters followed by 3 digits (8 no sequences of letters are prohibited, even if they are absence)

Ans:

$$26^3 \times 10^3$$

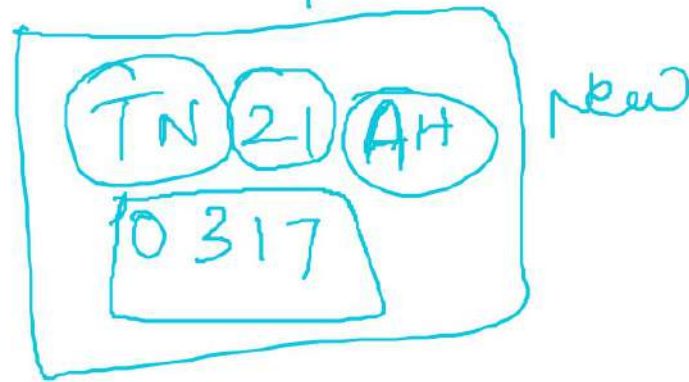
$A = \{n \text{ elements}\}$

$$|p(q)| \\ = 2^{|A|} \\ = 2^n$$

KPM

$|A| = n$

2016
- Jun



TAM
486

Tamil Nadu

Arahanam

Maigina (or) Month

TAM
444
~ ~ ~

TAA
~ ~

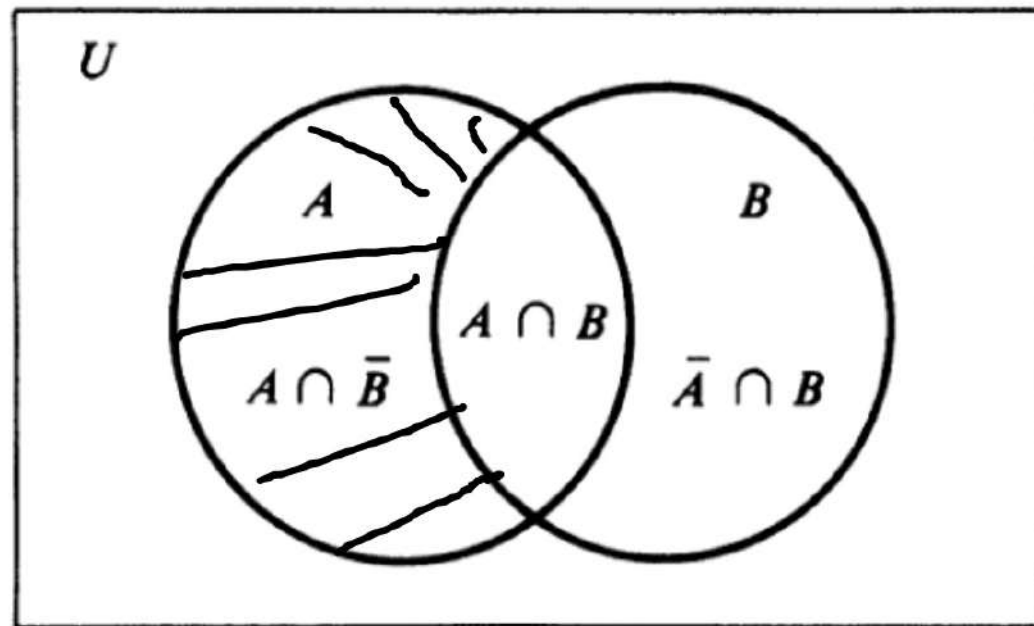
$$26 \times 26 \times 26 \times 10 \times 10 \times 10 \\ = 26^3 \times 10^3$$

Principle of Inclusion and Exclusion:

First Statement: If A and B are subsets of some universe set U , then

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

$$A \cap \bar{B} = A \setminus B$$



$$* |A \cup B| = \text{either } A \text{ or } B$$

$$= \{|A| + |B|\} - |A \cap B|$$

$$* |A \cup B \cup C| = \{|A| + |B| + |C|\} - \{|A \cap B| + |A \cap C| + |B \cap C|\} + |A \cap B \cap C|$$

$$* |A \cup B \cup C \cup D| = \{|A| + |B| + |C| + |D|\} - \{|A \cap B| + |A \cap C| + |A \cap D| + |B \cap C| + |B \cap D| + |C \cap D|\} + \{|A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D|\} - |A \cap B \cap C \cap D|$$

Theorem 2.8.1. If A , B , and C are finite sets, then

either A (or) B (or) C

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Example If there are 200 faculty members that speak French, 50 that speak Russian, 100 that speak Spanish, 20 that speak French and Russian, 60 that speak French and Spanish, 35 that speak Russian and Spanish, while only 10 speak French, Russian, and Spanish, how many speak either French or Russian or Spanish?

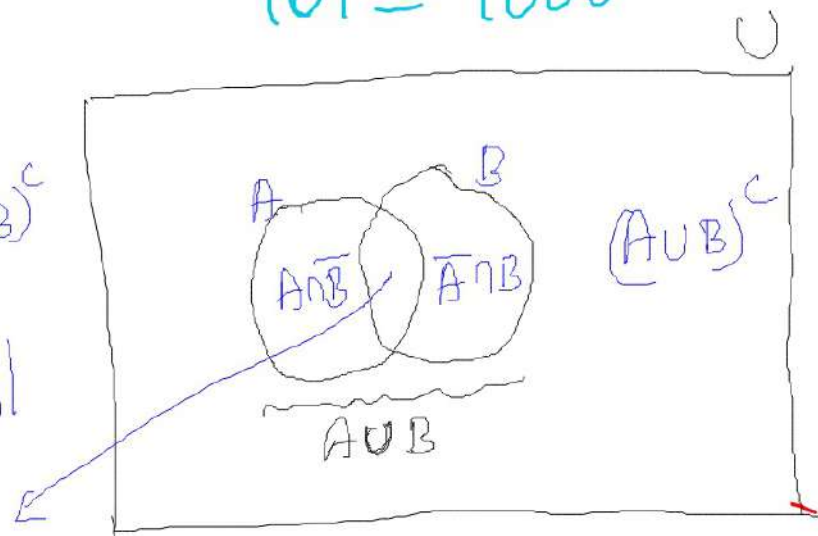
Solution: either F (or) R (or) S

Let F be the set of faculty who speak French, R be the set of faculty who speak Russian, and S be the set of faculty that speak Spanish. We know from example 2.8.1. that $|F| = 200$, $|R| = 50$, $|S| = 100$, $|F \cap R| = 20$, $|F \cap S| = 60$, $|R \cap S| = 35$ and $|F \cap R \cap S| = 10$. Thus, $|F \cup R \cup S| = 200 + 50 + 100 - 20 - 60 - 35 + 10 = 245$.

① No of faculty those who not speak
 either F (or) S (or) R = $|\overline{F \cup S \cup R}|$
 $= |U| - |F \cup S \cup R| = 1000 - 245 = 755$

✗ Let there are 1000 faculty members
 $|U| = 1000$

$U = (A \cup B)$
 $U (A \cup B)^c$
 $|U| = |A \cup B| + |(A \cup B)^c|$



$A \cap B$

① $|U| - |A \cup B|$
 $= |\overline{(A \cup B)}|$
 $= |\overline{A} \cap \overline{B}|$

✗ ① $|U| = |A \cup B| + |\overline{A \cup B}|$

$$|F \cup R \cup S| = \{ |F| + |R| + |S| \} - \{ |F \cap R| + |F \cap S| + |R \cap S| \} + |F \cap R \cap S|$$

Let $F \rightarrow$ Faculty who speak French
 $R \rightarrow$ " " " Russian
 $S \rightarrow$ " " " Spanish

$$|F| = 200, |R| = 50, |S| = 100, |F \cap R| = 20$$

$$|F \cap S| = 60, |R \cap S| = 35, |F \cap R \cap S| = 10$$

No. of faculty who speak either French
 (or) Russian (or) Spanish

$$= |F \cup R \cup S| = \{ 200 + 50 + 100 \} - \{ 20 + 60 + 35 \} + 10$$

$$= 350 - 115 + 10 = 360 - 115 = 245$$

Note: $| \overline{A \cup B} |$

$$\begin{aligned}
 | \overline{A} \cap \overline{B} | &= | \overline{A \cup B} | = | U | - | A \cup B | \\
 &= | U | - \{ | A | + | B | - | A \cap B | \} \\
 &= | U | - | A | - | B | + | A \cap B |.
 \end{aligned}$$

$| \text{Not } A \& \text{ Not } B |$

Example 2.8.4. If in Example 2.8.1 there are 1,000 faculty altogether, then there are $1,000 - | F | - | R | + | F \cap R | = 1,000 - 200 - 50 + 20 = 1,000 - | F \cup R | = 1,000 - 230 = 770$ people who speak neither French nor Russian.

Note:

$\text{not } (\text{either French (or) Spanish (or) Russian) }$
~~not~~ $(| F \cup B \cup R |)$

$$\begin{aligned}
 | \overline{A} \cap \overline{B} \cap \overline{C} | &= | \overline{A \cup B \cup C} | = | U | - | A \cup B \cup C | \\
 &= | U | - \{ | A | + | B | + | C | - | A \cap B | \\
 &\quad - | A \cap C | - | B \cap C | + | A \cap B \cap C | \} \\
 &= | U | - | A | - | B | - | C | + | A \cap B | \\
 &\quad + | A \cap C | + | B \cap C | - | A \cap B \cap C |.
 \end{aligned}$$

Example 2.8.5. In a survey of students at Florida State University the following information was obtained: 260 were taking a statistics course, 208 were taking a mathematics course, 160 were taking a computer programming course, 76 were taking statistics and mathematics, 48 were taking statistics and computer programming, 62 were taking mathematics courses and computer programming, 30 were taking all 3 kinds of courses, and 150 were taking none of the 3 courses.

Let

$$|U| - |S \cup M \cup C| = |\overline{S \cup M \cup C}| = 150$$

S = {students taking statistics},

M = {students taking mathematics},

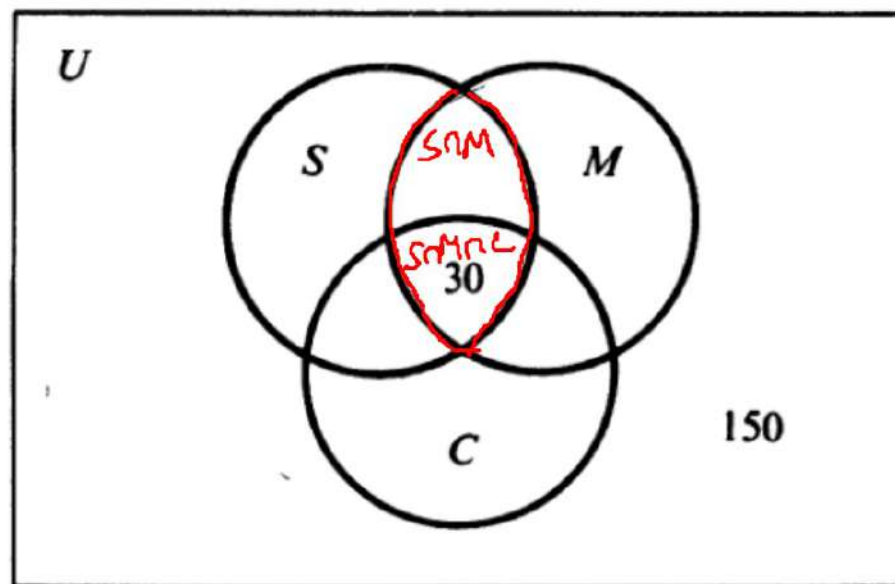
C = {students taking computer programming}.

- (a) How many students were surveyed? $|U| = |S \cup M \cup C| + |\overline{S \cup M \cup C}|$
- (b) How many students were taking a statistics and a mathematics course but not a computer programming course? $|S \cap M \cap \overline{C}|$
- (c) How many were taking a statistics and a computer course but not a mathematics course?

$$\overline{M} \quad |S \cap C \cap \overline{M}| = |S \cap C| - |S \cap C \cap M|$$

- (d) How many were taking a computer programming and a mathematics course but not a statistics course? $|C \cap M \cap \bar{S}|$
- (e) How many were taking a statistics course but not taking a course in mathematics or in computer programming? $|S \cap \bar{M} \cap \bar{C}|$
- (f) How many were taking a mathematics course but not taking a statistics course or a computer programming course? $|M \cap \bar{S} \cap \bar{C}|$
- (g) How many were taking a computer programming course but not taking a course in mathematics or in statistics? $|C \cap \bar{M} \cap \bar{S}|$

Solution:



$$\begin{aligned}
 & |S \cap \bar{M} \cap \bar{C}| \\
 &= |S| \\
 &\quad - |S \cap \bar{M} \cap C| \\
 &\quad - |S \cap M \cap \bar{C}| \\
 &\quad - |S \cap M \cap C|
 \end{aligned}$$

Figure 2-11

We know that $|S| = 260$, $|M| = 208$, $|C| = 160$, $|S \cap M| = 76$, $|S \cap C| = 48$, $|M \cap C| = 62$, $|S \cap M \cap C| = 30$, and $|\overline{S \cup M \cup C}| = 150$; so we can immediately insert the number of students in 2 of the 8 regions of the Venn diagram, the regions $S \cap M \cap C$ and $\overline{S \cup M \cup C}$.

(a) The total number of students surveyed:

$$\begin{aligned}
 |U| &= |S \cup M \cup C| + |\overline{S \cup M \cup C}| \\
 &= \{ |S| + |M| + |C| \} - |S \cap M| - |S \cap C| - |M \cap C| \\
 &\quad + |S \cap M \cap C| + |\overline{S \cup M \cup C}| \\
 &= \underline{260 + 208 + 160} - \underline{76} - \underline{48} - \underline{62} + \underline{30} + 150 = \underline{622}.
 \end{aligned}$$

(b) We are asked to find:

$$\begin{aligned}
 |S \cap M \cap \overline{C}| &= |S \cap M - S \cap M \cap C| = |S \cap M| - |S \cap M \cap C| \\
 &= 76 - 30 = 46.
 \end{aligned}$$

$$(c) |S \cap C \cap \overline{M}| = |S \cap C| - |S \cap C \cap M| = 48 - 30 = 18.$$

$$(d) |M \cap C \cap \overline{S}| = |M \cap C| - |M \cap C \cap S| = 62 - 30 = 32.$$

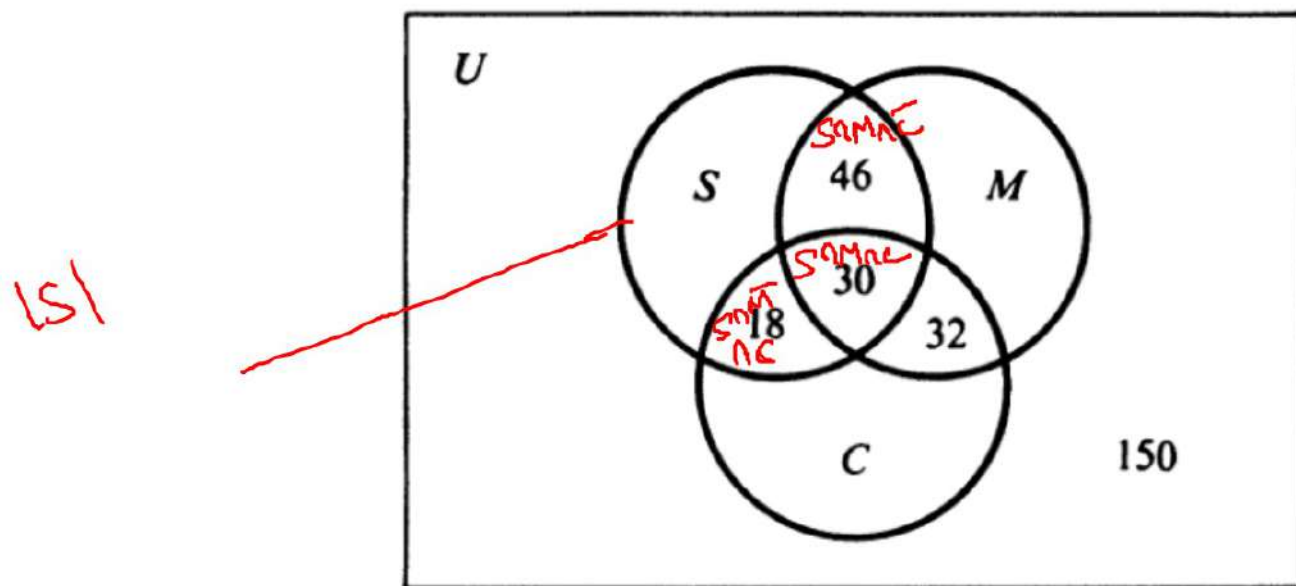


Figure 2-12

$$(e) |S \cap \bar{M} \cap \bar{C}| = |S| - |S \cap M \cap \bar{C}| - |S \cap C \cap \bar{M}| - |S \cap M \cap C| = 260 - 46 - 18 - 30 = 166.$$

$$(f) |M \cap \bar{S} \cap \bar{C}| = |M| - |M \cap S \cap \bar{C}| - |M \cap C \cap \bar{S}| - |M \cap S \cap C| = 100$$

$$(g) |C \cap \bar{S} \cap \bar{M}| = |C| - |C \cap S \cap \bar{M}| - |C \cap \bar{S} \cap M| - |C \cap S \cap M| = 80$$

Theorem 2.8.1. (General statement of the principle of inclusion-exclusion). If A_i are finite subsets of a universal set U , then

$$\begin{aligned}
 |A_1 \cup A_2 \cup \dots \cup A_n| = & \sum_{i=1}^n |A_i| - \sum_{i,j} |A_i \cap A_j| \\
 & + \sum_{i,j,k} |A_i \cap A_j \cap A_k| \\
 & + \dots + \\
 & (-1)^{n-1} |A_1 \cap A_2 \dots \cap A_n|,
 \end{aligned} \tag{2.8.10}$$

Note:

$$\begin{aligned}
 |A_1 \cup A_2 \cup A_3 \cup A_4| = & \left\{ |A_1| + |A_2| + |A_3| + |A_4| \right\} \\
 & - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_1 \cap A_4| \\
 & - |A_2 \cap A_3| - |A_2 \cap A_4| - |A_3 \cap A_4| \\
 & + |A_1 \cap A_2 \cap A_3| + |A_1 \cap A_2 \cap A_4| \\
 & + |A_1 \cap A_3 \cap A_4| + |A_2 \cap A_3 \cap A_4| \\
 & - |A_1 \cap A_2 \cap A_3 \cap A_4|.
 \end{aligned}$$

* To find:

The no/- of integers that are
divisible by either 2 (or) 3 (or) 5 (or) 7.

$$|(A \cup B \cup C \cup D)| = ?$$

Let $|A| \rightarrow$ no/- of integers that are
divisible by 2

$|B| \rightarrow$ no/- of integers " " 3

$|C| \rightarrow$ " " 5

$|D| \rightarrow$ " " 7

$|A \cap B| \rightarrow$ " " " 2 & 3

$$|A| = \left\lfloor \frac{1000}{2} \right\rfloor = \lfloor 500 \rfloor = 500$$

$$|B| = \left\lfloor \frac{1000}{3} \right\rfloor = \lfloor 333.33 \rfloor = 333$$

$$|C| = \left\lfloor \frac{1000}{5} \right\rfloor = \lfloor 200 \rfloor = 200$$

$$|D| = \left\lfloor \frac{1000}{7} \right\rfloor = \lfloor 142.85 \rfloor = 142$$

$$\begin{aligned} |A \cup B \cup C \cup D| &= |A| + |B| + |C| + |D| \\ &\quad - \{ |A \cap B| + |A \cap C| + |A \cap D| \\ &\quad \quad + |B \cap C| + |B \cap D| + |C \cap D| \} \\ &\quad + \{ |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| \\ &\quad \quad + |B \cap C \cap D| \} \\ &\quad - |A \cap B \cap C \cap D| \end{aligned}$$

$$|A \cap B| = \left\lfloor \frac{1000}{6} \right\rfloor = \lfloor 166.66 \rfloor = 166$$

$$|A \cap C| = \left\lfloor \frac{1000}{10} \right\rfloor = \lfloor 100 \rfloor = 100$$

$$|A \cap D| = \left\lfloor \frac{1000}{14} \right\rfloor = \lfloor 71.42 \rfloor = 71$$

$$|B \cap C| = \left\lfloor \frac{1000}{15} \right\rfloor = \lfloor 66.66 \rfloor = 66$$

$$|B \cap D| = \left\lfloor \frac{1000}{21} \right\rfloor = \lfloor 47.61 \rfloor = 47$$

$$|C \cap D| = \left\lfloor \frac{1000}{35} \right\rfloor = \lfloor 28.57 \rfloor = 28$$

$$|A \cap B \cap C| = \left\lfloor \frac{1000}{2 \times 3 \times 5} \right\rfloor = \left\lfloor \frac{1000}{30} \right\rfloor = \left\lfloor \frac{100}{3} \right\rfloor = \lfloor 33.33 \rfloor = 33$$

$$|A \cap B \cap D| = \left\lfloor \frac{1000}{2 \times 3 \times 7} \right\rfloor = \left\lfloor \frac{1000}{42} \right\rfloor = \lfloor 23.80 \rfloor = 23$$

$$|A \cap C \cap D| = \left\lfloor \frac{1000}{2 \times 5 \times 7} \right\rfloor = \left\lfloor \frac{1000}{70} \right\rfloor = \lfloor 14.28 \rfloor = 14$$

$$|B \cap C \cap D| = \left\lfloor \frac{1000}{3 \times 5 \times 7} \right\rfloor = \left\lfloor \frac{1000}{105} \right\rfloor = \lfloor 9.52 \rfloor = 9$$

$$|A \cap B \cap C \cap D| = \left\lfloor \frac{1000}{2 \times 3 \times 5 \times 7} \right\rfloor = \left\lfloor \frac{1000}{210} \right\rfloor = \lfloor 4.76 \rfloor = 4$$

No. of integers that are divisible by
either 2 (or) 3 (or) 5 (or) 7

$$= |A \cup B \cup C \cup D|$$

$$= \{ |A| + |B| + |C| + |D| \} - \{ |A \cap B| + |A \cap C| + |A \cap D| \\ + |B \cap C| + |B \cap D| \\ + |C \cap D| \}$$

$$+ \{ |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D| \} \\ - |A \cap B \cap C \cap D|$$

$$= \{ 500 + 333 + 200 + 142 \} - \{ 166 + 100 + 71 \\ + 66 + 47 \\ + 28 \} \\ + \{ 33 + 23 + 14 + 9 \} - 4$$

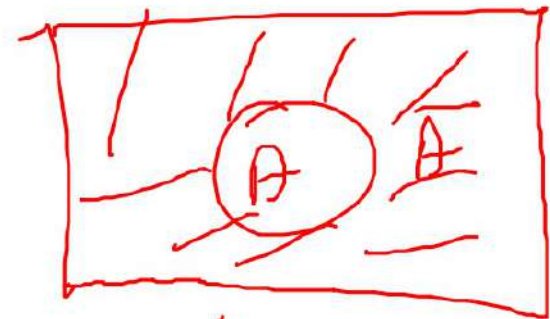
$$= 1175 - 478 + 79 - 4 = 1254 - 482 \\ = 772$$

The no. of integers that are not divisible
by either 2 (or) 3 (or) 5 (or) 7

$$= |\overline{A \cup B \cup C \cup D}| = |U| - |A \cup B \cup C \cup D|$$

$$= 1000 - 772$$

$$= 228$$



$$A \cup \bar{A} = U$$

$$\bar{A} = U - A$$

$$A = U - \bar{A}$$

Problem : 2

How many integers are there between 1 and 1000 both inclusive that are not divisible by 2, 3, 5 and 7?

Solution:

Let U be the set of integers x such that $1 \leq x \leq 1,000$. Let A_1, A_2, A_3, A_4 be the set of elements of U divisible by 2, 3, 5, and 7, respectively. Thus, $A_1 \cap A_2$ denotes those positive integers $\leq 1,000$ that are divisible by 6, $A_1 \cap A_3$ those divisible by 10, $A_2 \cap A_3 = \{\text{all integers } x \mid 1 \leq x \leq 1,000 \text{ and } x \text{ is divisible by 15}\}$ and so on. We wish to compute $|\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap \bar{A}_4|$. We know that

$$|A_1| = \frac{1,000}{2} = 500, \quad |A_2| = \left\lfloor \frac{1,000}{3} \right\rfloor$$

(where $\lfloor x \rfloor$ means the greatest integer $\leq x$),

$$|A_3| = \left\lfloor \frac{1,000}{5} \right\rfloor = 200,$$

$$\begin{aligned} & |\overline{A_1 \cup A_2 \cup A_3 \cup A_4}| \\ &= |\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap \bar{A}_4| \end{aligned}$$

$$|A_4| = \left\lfloor \frac{1,000}{7} \right\rfloor = 142,$$

$$|A_1 \cap A_2| = \left\lfloor \frac{1,000}{6} \right\rfloor = 166$$

$$|A_1 \cap A_3| = \frac{1,000}{10} = 100,$$

$$|A_1 \cap A_4| = \left\lfloor \frac{1,000}{14} \right\rfloor = 71,$$

$$|A_2 \cap A_3| = \left\lfloor \frac{1,000}{15} \right\rfloor = 66,$$

$$|A_2 \cap A_4| = \left\lfloor \frac{1,000}{21} \right\rfloor = 47,$$

$$|A_3 \cap A_4| = \left\lfloor \frac{1,000}{35} \right\rfloor = 28,$$

$$|A_1 \cap A_2 \cap A_3| = \left\lfloor \frac{1,000}{30} \right\rfloor = 33,$$

$$|A_1 \cap A_2 \cap A_4| = \left\lfloor \frac{1,000}{42} \right\rfloor = 23,$$

$$|A_1 \cap A_3 \cap A_4| = \left\lfloor \frac{1,000}{70} \right\rfloor = 14,$$

$$|A_2 \cap A_3 \cap A_4| = \left\lfloor \frac{1,000}{105} \right\rfloor = 9,$$

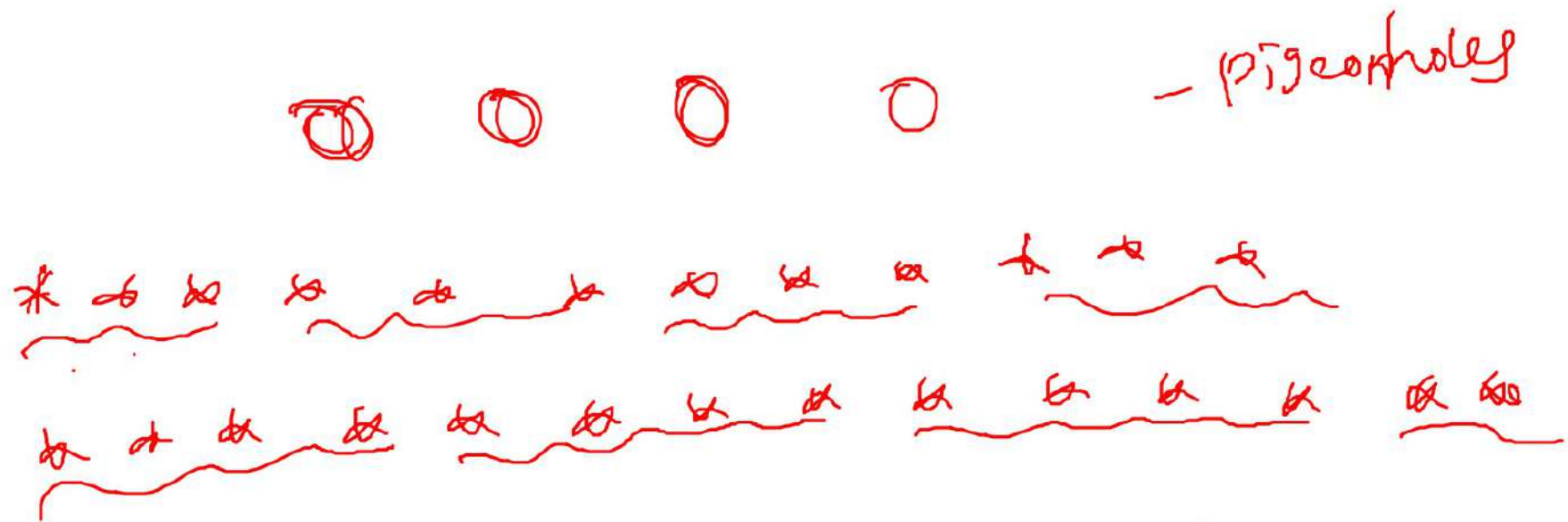
$$|A_1 \cap A_2 \cap A_3 \cap A_4| = \left\lfloor \frac{1,000}{210} \right\rfloor = 4.$$

Then,

$$\begin{aligned}|A_1 \cup A_2 \cup A_3 \cup A_4| &= 500 + 333 + 200 + 142 - 166 \\ &\quad - 100 - 71 - 66 - 47 - 28 \\ &\quad + 33 + 23 + 14 + 9 - 4 = 772.\end{aligned}$$

Thus,

$$|\overline{A_1} \cap \overline{A_2} \cap \overline{A_3} \cap \overline{A_4}| = 1,000 - 772 = 228.$$



n - pigeons
 m - pigeon holes $(n > m)$

At least
 one hole
 containing
 $\left\lfloor \frac{n-1}{m} \right\rfloor + 1$ pigeons can be
 accommodated in any one of the
 pigeon hole.

The pigeon hole principle

Statement

If n pigeons fly into m pigeonholes, then (at least) one pigeonhole contains (at least)

$$\lceil \frac{n}{m} \rceil \text{ pigeons. (or) } \left\lfloor \frac{n-1}{m} \right\rfloor + 1 \text{ pigeons.}$$

where $n > m$

Example

① Suppose that 27 pigeons fly into a coop with 5 pigeon holes. What is the maximum number m that could fly into any single pigeon hole?

Sol Given $\lceil \frac{n}{m} \rceil = \lceil \frac{27}{5} \rceil = \lceil 5.4 \rceil = 6$

At least 6 pigeons can be accommodated in one pigeon hole.

$$h=27, m=5$$

$$^5_0 \quad ^{0_5}_5 \quad ^{0_5}_6 \quad ^{0_6}_6$$

$$\alpha \quad \left\lfloor \frac{n-1}{m} \right\rfloor + 1 = \left\lfloor \frac{27-1}{5} \right\rfloor + 1 = \left\lfloor \frac{26}{5} \right\rfloor + 1$$

$$= \lfloor 5 \cdot 2 \rfloor + 1 = 5 + 1 = 6$$

$$\ast \quad \lceil n/m \rceil = \lceil 27/5 \rceil = \lceil 5.4 \rceil = 6$$

② If seven colours are used to paint 50 cars then find ^{atleast} how many cars will have same colour.

Sol Given $n = 50$, $m = 7$

$$\left\lceil \frac{n}{m} \right\rceil = \left\lceil \frac{50}{7} \right\rceil = \lceil 7.1 \rceil = 8 \text{ Cars}$$

$\therefore$ Atleast 8 Cars will have same colour.

③ If 7 members of a family have total Rs. 2886 in their pockets then find at least how much amount will the person have in their pocket?

Sol

$$n = 7 \text{ \& } m = 2886$$

$$\left\lceil \frac{n}{m} \right\rceil = \left\lceil \frac{2886}{7} \right\rceil = \text{Rs. } 413$$

$\therefore$ At least Rs. 413 in his pocket.

④. How many people must you have to guarantee that atleast 9 of them will have birthdays in the same day of the week.

Sol

Case: 1 $n = ?$
 $m = 7$

Given $\left\lceil \frac{n}{m} \right\rceil = 9$

$$\frac{n}{7} = 9$$

$$n = 63 \text{ people}$$

(or)

Case: 2
 $n = ?$, $m = 7$

$$\left\lfloor \frac{n-1}{m} \right\rfloor + 1 = 9$$

$$\frac{n-1}{7} = 9-1$$

$$n-1 = 56$$

$$n = 57 \text{ people.}$$

Given Atleast 9 of them will have same birthday in a week.

to find: No. of people? ($n=?$)
Formula

Atleast $\left\lfloor \frac{n-1}{m} \right\rfloor + 1$ Pigeons can be accommodated in any one of the hole.

$$\left\lfloor \frac{n-1}{m} \right\rfloor + 1 = 9, \quad m = 7$$

$$\left\lfloor \frac{n-1}{7} \right\rfloor + 1 = 9$$

$$\frac{n-1}{7} = 8$$

$$n-1 = 56$$

$$n = 57$$

$n=57$
 $\begin{matrix} 0 \\ 9 \\ \text{f} \end{matrix}$
 $\begin{matrix} 0 \\ \text{f} \end{matrix}$ (or) $\begin{matrix} 0 \\ 9 \\ \text{f} \end{matrix}$ (or) $\begin{matrix} 0 \\ 9 \\ \text{f} \end{matrix}$ (or) $\begin{matrix} 0 \\ 9 \\ \text{f} \end{matrix}$ (or) $\begin{matrix} 0 \\ 9 \\ \text{f} \end{matrix}$ (or) $\begin{matrix} 0 \\ 9 \\ \text{f} \end{matrix}$

Atleast

$\lceil \frac{n}{m} \rceil$ pigeons can be accommodated in any one of the holes.

$$\lceil \frac{n}{m} \rceil = 9$$

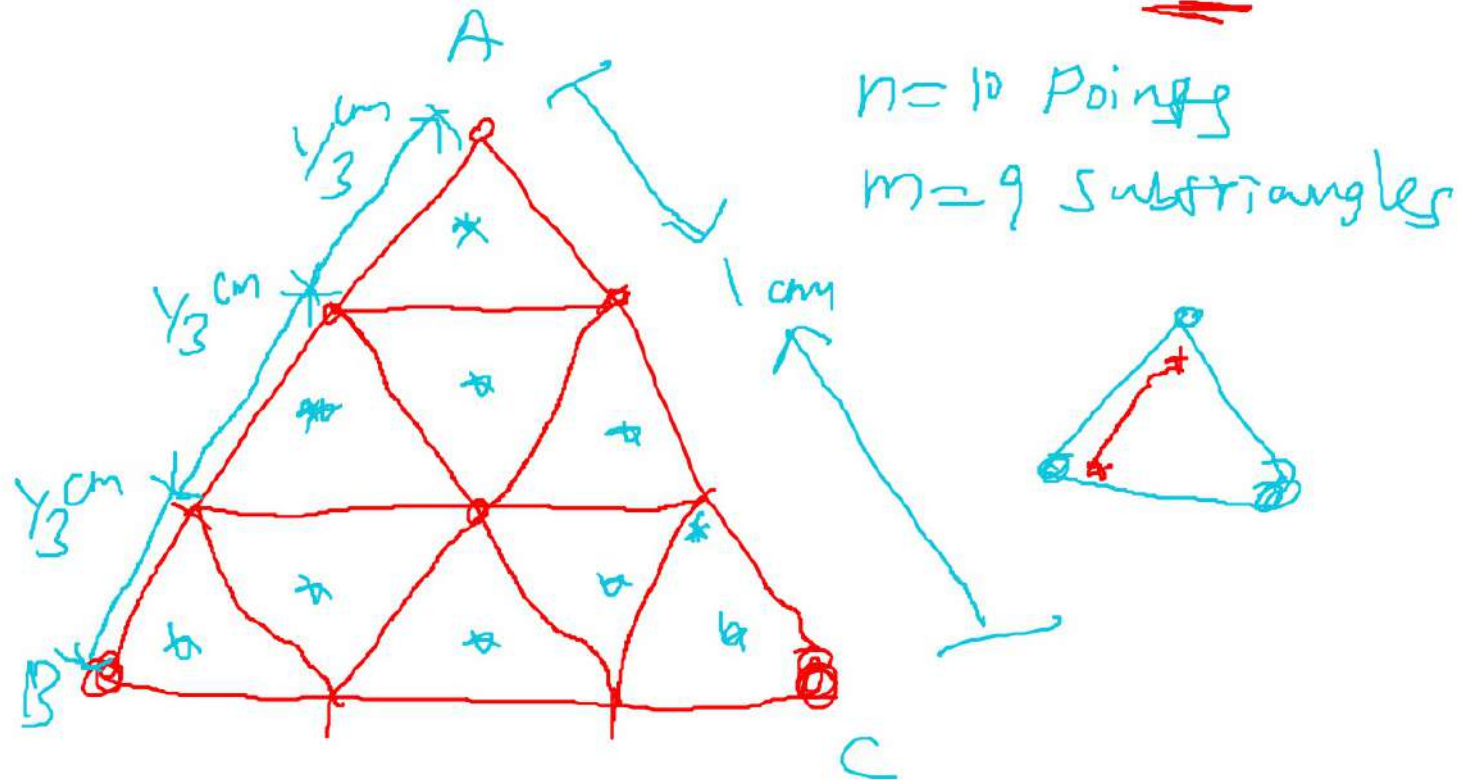
$$\lceil \frac{n}{4} \rceil = 9$$

$$n = 63$$

$$57 \leq n \leq 63$$

① $\triangle ABC$ is an equilateral triangle of side 1 cm each. If we select 10 points inside the triangle p.t. at least two points are there with distance less than $\frac{1}{3}$ cm.

Proof



Permutation and Combination:

Example:

In how many ways can we select three students from a group of five students to stand in line for a picture? In how many ways can we arrange all five of these students in a line for a picture?

Solution: First, note that the order in which we select the students matters. There are five ways to select the first student to stand at the start of the line. Once this student has been selected, there are four ways to select the second student in the line. After the first and second students have been selected, there are three ways to select the third student in the line. By the product rule, there are $5 \cdot 4 \cdot 3 = 60$ ways to select three students from a group of five students to stand in line for a picture.

To arrange all five students in a line for a picture, we select the first student in five ways, the second in four ways, the third in three ways, the fourth in two ways, and the fifth in one way. Consequently, there are $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ ways to arrange all five students in a line for a picture. 

A **permutation** of a set of distinct objects is an ordered arrangement of these objects. We also are interested in ordered arrangements of some of the elements of a set. An ordered arrangement of r elements of a set is called an **r -permutation**.

EXAMPLE 2 Let $S = \{1, 2, 3\}$. The ordered arrangement 3, 1, 2 is a permutation of S . The ordered arrangement 3, 2 is a 2-permutation of S . 

The number of r -permutations of a set with n elements is denoted by $P(n, r)$. We can find $P(n, r)$ using the product rule.

Example:

Let $S = \{a, b, c\}$. The 2-permutations of S are the ordered arrangements a, b ; a, c ; b, a ; b, c ; c, a ; and c, b . Consequently, there are six 2-permutations of this set with three elements. There are always six 2-permutations of a set with three elements. There are three ways to choose the first element of the arrangement. There are two ways to choose the second element of the arrangement, because it must be different from the first element. Hence, by the product rule, we see that $P(3, 2) = 3 \cdot 2 = 6$. the first element. By the product rule, it follows that $P(3, 2) = 3 \cdot 2 = 6$. 

Theorem:

If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$$

r -permutations of a set with n distinct elements.

$$S = \{a, b, c\}$$

$$|S| = 3$$

AB

BA

ab, ba, ac, bc, ca, cb

*

$3P_2$

— out of 3

6 arrangements

arranging 2 at a time

(ab $\neq$ ba)

Permutation

— order is a matter

n-objects

To

arrange : r-objects out of n objects

Combination

— order is not a

matter

(ab = ba)

$${}^n P_r \text{ (or) } n P_r$$

$$= \frac{n!}{(n-r)!}$$

$$n! = n \times (n-1)(n-2) \dots \dots 3 \cdot 2 \cdot 1$$

$$= n(n-1) \dots (n-(r+1)) \dots 3 \cdot 2 \cdot 1$$

$$= n(n-1) \dots (n-(r-1)) (n-r) \\ (n-(r+1)) \dots 3 \cdot 2 \cdot 1$$

$$= n(n-1) \dots (n-r+1) (n-r) (n-r-1) \\ \dots 3 \cdot 2 \cdot 1$$

$$nP_r = \frac{n!}{(n-r)!}$$

$$= \frac{n(n-1) \dots (n-(r-1)) \cancel{(n-r)} \cancel{(n-(r+1))} \dots 3 \cdot 2 \cdot 1}{\cancel{(n-r)} \cancel{(n-(r+1))} \dots 3 \cdot 2 \cdot 1}$$

$$nPr = n(n-1)(n-2) \dots (n-(r-1))$$

$$\underline{7P_4} = \underline{7 \times 6 \times 5 \times 4} = 840$$

$$\underline{6P_3} = 6 \times 5 \times 4 = 120$$

$$nPr = \frac{n!}{(n-r)!}$$

$$7P_4 = \frac{7!}{(7-4)!} = \frac{7!}{3!} = \frac{7 \times 6 \times 5 \times 4 \times \cancel{3 \times 2 \times 1}}{\cancel{3 \times 2 \times 1}}$$

$$0! = 1$$

$$1! = 1$$

$$\left. \begin{array}{l} 0! = 1 \\ 1! = 1 \end{array} \right\} \begin{array}{l} nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} \\ = n! \end{array}$$

begin with c & end with a

$$\underline{c} \quad \underline{b, d, e, f, g} \quad \underline{a}$$
$$1P_1 \quad * \quad 5P_5 \quad + \quad 1P_1$$

$$1! \times 5! \times 1! = 1 \times 5 \times 4 \times 3 \times 2 \times 1 \times 1$$
$$= 120$$

* c & a always occupy the end places

$$\underline{b, d, e, f, g} \quad \underline{c \quad a}$$
$$5P_5 \times 2P_2 = 5! \times 2!$$
$$= 120 \times 2 = 240$$

③ How many permutations of the letters ABCDEFG contain a) the string BCD, b) the string CFGA c) the string BA & GF d) the strings ABC & DE e) the strings ABC & CDE?

Sol

Given A, B, C, D, E, F, G - 7 letters

a) A, BCD, E, F, G - 5 letters

$$5 \times 4 \times 3 \times 2 \times 1$$

$${}^5P_5 = 5! = 120$$

b) B, CFGA, D, E - 4 letters

$${}^4P_4 = 4! = 24 \text{ ways}$$

c) BA & GF

BA, C, D, E, GF - 5 letters

$$5P_5 = 5! = 120 \text{ ways}$$

d)

ABC & DE

ABC, DE, F, G - 4 letters

$$4P_4 = 4! = 24 \text{ ways}$$

e)

ABC, CDE = ABCDE, F, G - 3 letters

$$3P_3 = 3! = 6 \text{ ways}$$

Note: ^x
ABC, EFC

0 permutation.

* Discrete mathematics - T. Veerayajan
 For extra problems

$$\begin{aligned}
 nC_r &= \frac{n!}{r! (n-r)!} = \frac{n(n-1) \dots (n-(r-1)) \cancel{(n-r)} \dots \cancel{(n-(r+1))}}{\underbrace{r \times (r-1) \dots 3 \times 2 \times 1}_{\text{--- } 3 \times 2 \times 1} \times \underbrace{(n-r) \dots (n-(r+1))}_{\text{--- } 3 \times 2 \times 1}} \\
 &= \frac{n(n-1) \dots (n-(r-1))}{r(r-1) \dots 3 \cdot 2 \cdot 1}
 \end{aligned}$$

Note:



① If $nC_x = nC_y$

then either $n = x+y$ (or) $x=y$

② $nC_k = nC_{n-k}$

③ Combination (or) Number of k -Subsets of
an n -Set.

④

n -Set — Set containing ' n ' elements.

⑤ k -Subsets — Each subset k elements.

Unordered — $ab = ba$ — Combination
ordered — $ab \neq ba$ — Permutation

① $\{1, 2, 3, 4\}$ — 4 set

$\{1, 2, 3\}$, $\{2, 3, 4\}$, $\{1, 3, 4\}$
 $\{1, 2, 4\}$ (3 subsets)
 ~~$\{3, 2, 1\}$~~

$$4C_3 = 4C_{4-3} = 4C_1 = 4$$

$$4C_3 = \frac{4 \times 3 \times 2}{3 \times 2 \times 1} = \frac{4}{1} = 4$$

set = {a, b, c, d, e}

|2-subsets| = |{a, b}, {a, c}, {a, d}, {a, e}, {b, c}, {b, d}, {b, e}, {c, d}, {c, e}, {d, e}|

ab ac ad ae }
~~ba~~ bc bd be }
cd ce }
de } 10

$${}^5C_2 = \frac{5 \times 4}{2 \times 1} = 10$$

COROLLARY 1

If n and r are integers with $0 \leq r \leq n$, then $P(n, r) = \frac{n!}{(n-r)!}$.

Combination:

An **r -combination** of elements of a set is an unordered selection of r elements from the set.

Thus, an r -combination is simply a subset of the set with r elements.

Example-1: $\rightarrow$ 4-set $\rightarrow$ 3-Subsets

Let S be the set $\{1, 2, 3, 4\}$. Then $\{1, 3, 4\}$ is a 3-combination from S . (Note that $\{4, 1, 3\}$ is the same 3-combination as $\{1, 3, 4\}$, because the order in which the elements of a set are listed does not matter.)

The number of r -combinations of a set with n distinct elements is denoted by $C(n, r)$. Note that $C(n, r)$ is also denoted by $\binom{n}{r}$ and is called a **binomial coefficient**.

Example-2

We see that $C(4, 2) = 6$, because the 2-combinations of $\{a, b, c, d\}$ are the six subsets $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, $\{b, c\}$, $\{b, d\}$, and $\{c, d\}$.

Theorem:

The number of r -combinations of a set with n elements, where n is a nonnegative integer and r is an integer with $0 \leq r \leq n$, equals

$$C(n, r) = \frac{n!}{r!(n-r)!}.$$

Note:

$$C(n, r) = \frac{n!}{r!(n-r)!} = \frac{n(n-1) \cdots (n-r+1)}{r!}.$$

COROLLARY 2

Let n and r be nonnegative integers with $r \leq n$. Then $C(n, r) = C(n, n-r)$.

✓ $nCr = nC_{n-r}$ for $r \leq n$

Note: $10C_8 = 10C_{10-8} = 10C_2$
 $= \frac{10 \times 9}{2 \times 1} = 45$

Examples:

How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?

Solution: The answer is given by the number of 5-combinations of a set with 10 elements. By Theorem 2, the number of such combinations is

$$C(10, 5) = \frac{10!}{5!5!} = 252.$$

$$10C_5 = \binom{10}{5} = \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1} = 252$$

A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission (assuming that all crew members have the same job)?

Solution: The number of ways to select a crew of six from the pool of 30 people is the number of 6-combinations of a set with 30 elements, because the order in which these people are chosen does not matter. By Theorem 2, the number of such combinations is

$$C(30, 6) = \frac{30!}{6!24!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 593,775.$$

Suppose that there are 9 faculty members in the mathematics department and 11 in the computer science department. How many ways are there to select a committee to develop a discrete mathematics course at a school if the committee is to consist of three faculty members from the mathematics department and four from the computer science department?

Solution: By the product rule, the answer is the product of the number of 3-combinations of a set with nine elements and the number of 4-combinations of a set with 11 elements. By Theorem 2, the number of ways to select the committee is

$$9C_3 \times 11C_4$$

$$C(9, 3) \cdot C(11, 4) = \frac{9!}{3!6!} \cdot \frac{11!}{4!7!} = 84 \cdot 330 = 27,720.$$

TABLE 1 Combinations and Permutations With and Without Repetition.

Type	Repetition Allowed?	Formula
<i>r</i> -permutations	No	$\frac{n!}{(n-r)!}$
<i>r</i> -combinations	No	$\frac{n!}{r!(n-r)!}$
<i>r</i> -permutations	Yes	n^r
<i>r</i> -combinations	Yes	$\frac{(n+r-1)!}{r!(n-1)!}$

nPr

nCr

$n+r-1Cr$

① If ${}^{20}C_{\underline{n+2}} = {}^{20}C_{\underline{2n-1}}$ then find n ?

$$n = x + y$$

$$n = x + 2 + 2n - 1$$

$$0 = 1 + 2n$$

$$2n = -1$$

$$n = -\frac{1}{2} \text{ Not possible.}$$

($\therefore$ If $nCx = nCy$
then either
 $n = x + y$
(or) $x = y$)

$$\therefore x = y$$

$$n+2 = 2n - 1$$

$$\boxed{3 = n}$$

① How many bit strings of length 10 contain

a) exactly four 1's

b) At most four 1's

c) At least four 1's

d) An equal number of 0's & 1's

Sol

* Given a bit string of length 10

* So we have 10 positions.

a) Exactly four 1's

Four places are filled with ones

Remaining places are filled with zeros

$${}^{10}C_4 = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$$

b) At most four 'S'!

$${}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3 + {}^{10}C_4$$
$$= 1 + 10 + \frac{10 \times 9}{2 \times 1} + \frac{10 \times 9 \times 8}{3 \times 2 \times 1} + 210$$

$$= 1 + 10 + 45 + 120 + 210 = 386$$

$$\boxed{\begin{matrix} nCr \\ = nCr \end{matrix}}$$

c) At least four 'S'!

$${}^{10}C_4 + {}^{10}C_5 + {}^{10}C_6 + {}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9$$
$$+ {}^{10}C_{10}$$
$$= 210 + {}^{10}C_5 + {}^{10}C_4 + {}^{10}C_3 + {}^{10}C_2$$
$$+ {}^{10}C_1 + {}^{10}C_0$$

$$= 210 + \frac{10^3 \cancel{9} \times \cancel{8}^2 \times 7 \times 6}{\cancel{5} \times \cancel{4} \times \cancel{3} \times \cancel{2} \times 1} + \frac{10^3 \cancel{9} \times \cancel{8} \times 7}{\cancel{4} \times \cancel{3} \times \cancel{2} \times 1}$$

$$+ 120 + 45 + 10 + 1$$

$$= 210 + 36 \times 7 + 210 + 120 + 45 + 1$$

$$= 848$$

d) An even no/- of Zero's & One's

$${}^{10}C_5 = \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times \cancel{3} \times \cancel{2} \times 1} = 36 \times 7 = 252$$